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The University of Manchester

EART11100

How to Build a Planet

Week 4 An introduction to minerals



An interlude in the story of Earth-building

Topics

The first part of this lecture we will cover:

- the physical basis for the definition of a mineral and use this to distinguish minerals from non-minerals
- the different ways in which minerals form
- the five most common classes of rock-forming minerals with examples
- the formation of silicate minerals
- the main physical properties of minerals and how these could be used to distinguish minerals

Minerals

- **A mineral is a naturally occurring chemical substance which has a definite atomic structure and chemical formula.**
- **They are crystalline, solid inorganic materials.**
- Where minerals can grow freely and without constraint, e.g in a cavity, they will form well-formed crystals that reflect their atomic structure.
- There are about 5700 naturally occurring minerals on Earth.
- Webmineral.com and Mindat.org are excellent online databases containing information about all known minerals.
- Minerals are the building blocks of rocks.
- Rocks consist of an assemblage of minerals of 1 or more different types.

Minerals are natural materials



Real Diamonds



Synthetic Diamonds

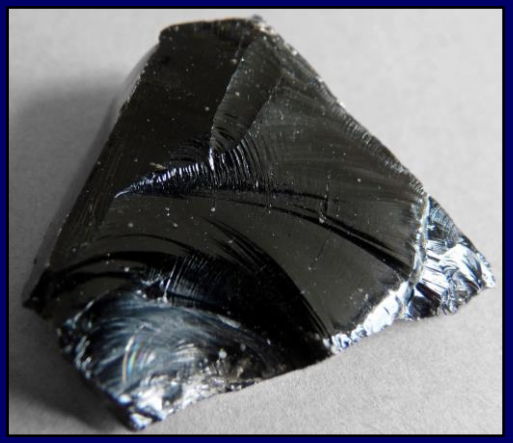




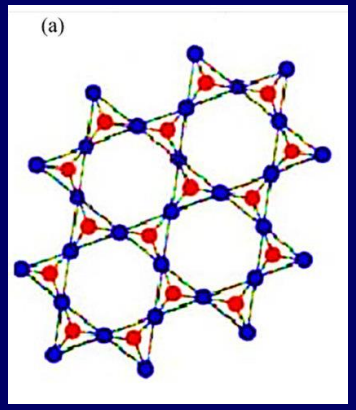
Minerals are solid and crystalline



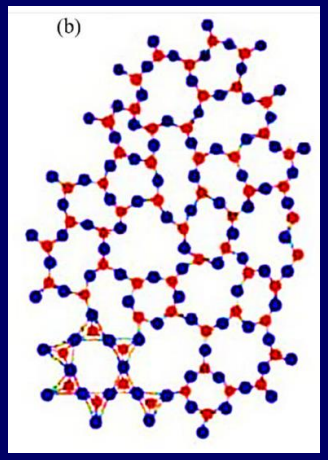
Quartz (SiO_2) the most abundantly occurring mineral on Earth



Natural and man-made glass is amorphous (without form) and is not a mineral







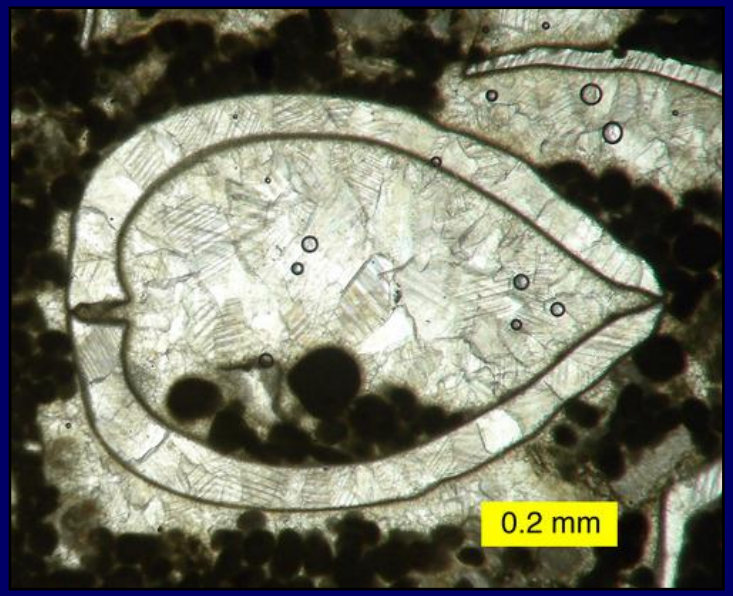
Minerals are inorganic



The carbon in coal is organic, and therefore coal is not a mineral ……



…. but graphite is a mineral



Calcite ($CaCO_3$) and aragonite are secreted by organisms to make their hard parts. But carbon in this structure is inorganic and therefore calcite and aragonite are minerals

Minerals have a specific composition



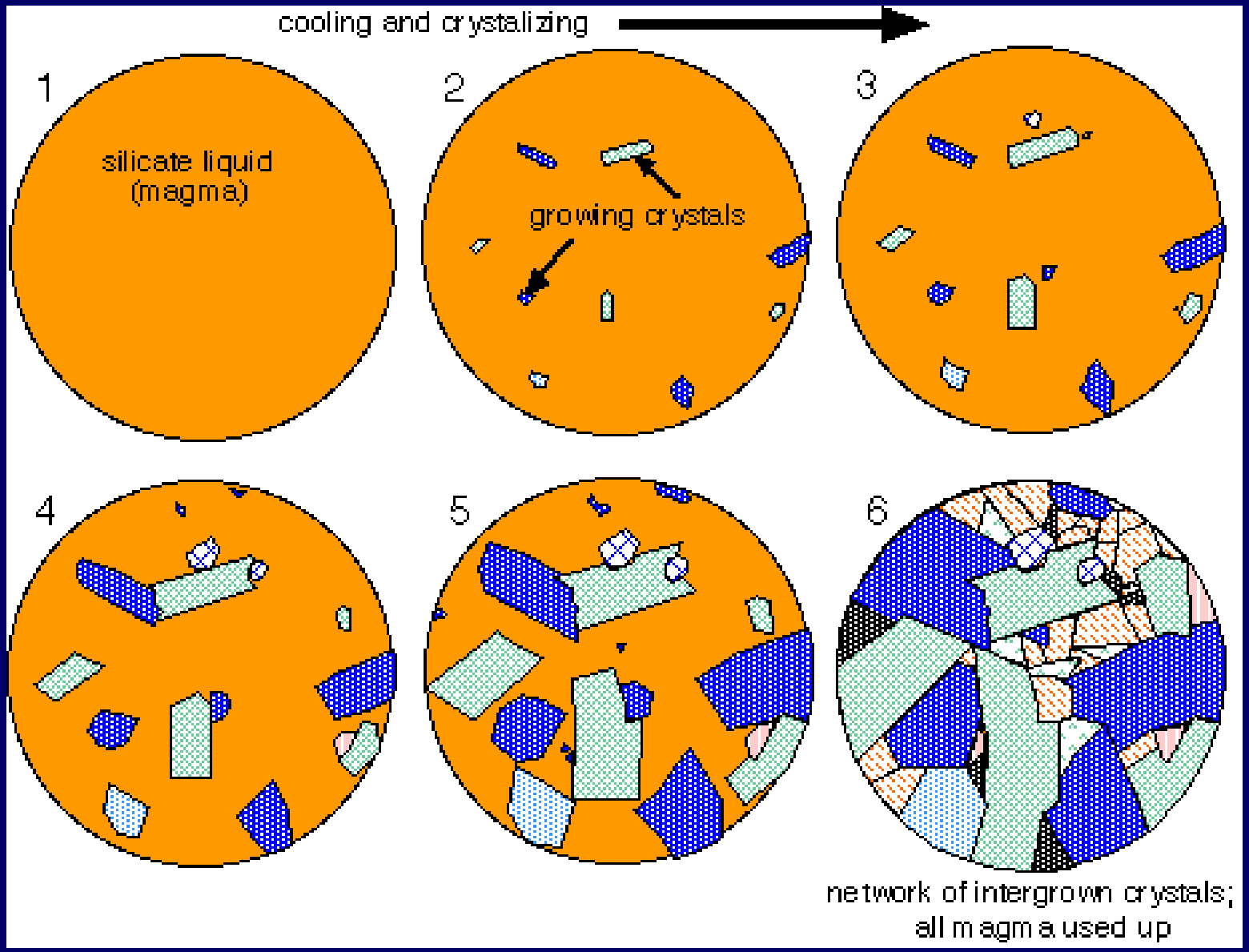


Quartz (SiO_2) has a fixed ratio of two atoms of oxygen to one atom of silicon



Olivine $(Mg, Fe)_2SiO_4$ has a fixed ratio of (iron + magnesium) atoms to silicon and oxygen atoms. The ratio of iron and magnesium varies.

The formation of minerals: 1. Cooling and crystallisation



The formation of minerals: 2. Evaporation and precipitation



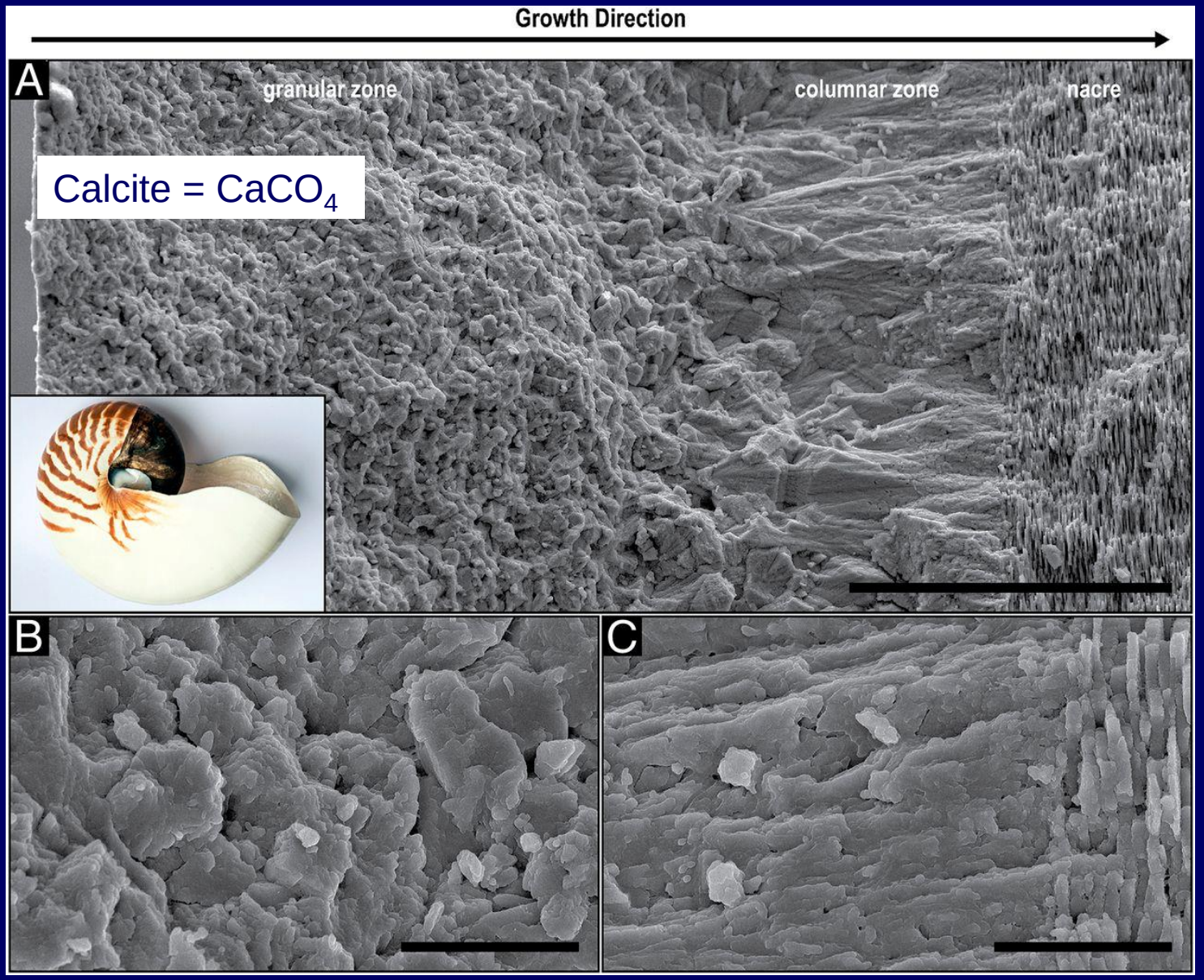
The formation of minerals: 3. Sold state transformation



15 GPa

1300 K

The formation of minerals: 4. Secretion by organisms



Classes of Rock-Forming Minerals

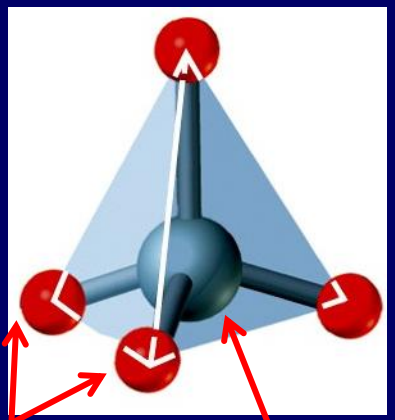
Silicates – contain O and Si e.g., quartz (SiO_2)

Carbonates – contain CO_3^{2-} e.g., calcite ($CaCO_3$)
Oxides – contain O^{2-} and metallic cations e.g., hematite (Fe_2O_3)
Sulfides – contain S^- (but not oxygen) and metallic cations e.g., pyrite (FeS_2)
Sulfates – contain SO_4^{2-} and metallic cations e.g., anhydrite ($CaSO_4$)

Halides – contain one of Cl^-, F^-, I^-, or Br^- (but no oxygen)
Hydroxides – contain OH^-
Native elements – metals occurring as a single element, not in combination e.g., gold, silver , copper

Formation of silicate minerals

Silicate ion (SiO_4^{4-})



Oxygen ions (O^{2-})

Silicon ion (Si^{4+})

% crustal abundances

Silicon (Si): 46.6%
Oxygen (O): 27.7%

Negative charge

Can bond with cations to form minerals

- Aluminium (Al): 8.1%
- Iron (F): 5.0%
- Calcium (Ca): 3.6%
- Sodium (Na): 2.8%
- Potassium (K): 2.6%
- Magnesium (Mg): 2.1%
- All others: 1.5%

Crustal abundance

Silicate tetrahedra are the building blocks of silicates minerals – polymerisation

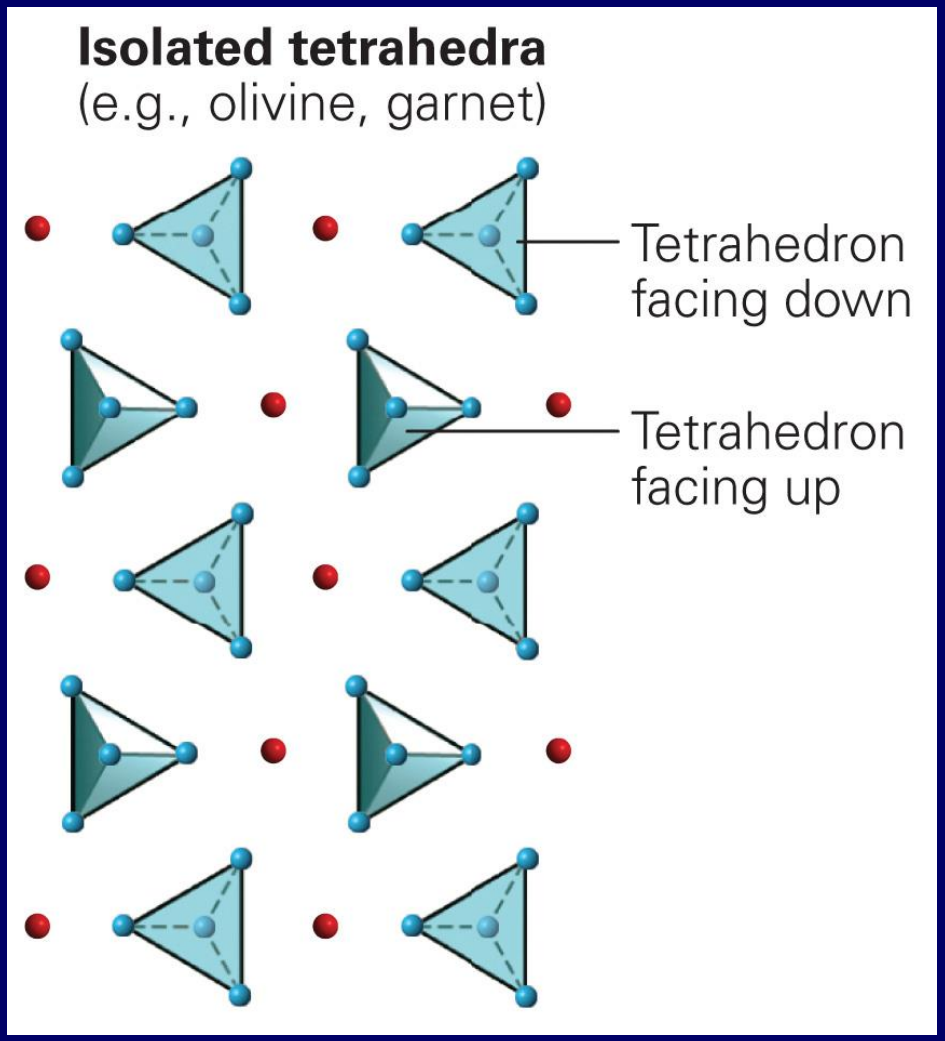
Formation of silicate minerals: Islands



Olivine $(Mg, Fe)_2SiO_4$



Almandine Garnet $Fe_3Al_2(SiO_4)_3$



Silicate minerals with isolated tetrahedra do not share any oxygens

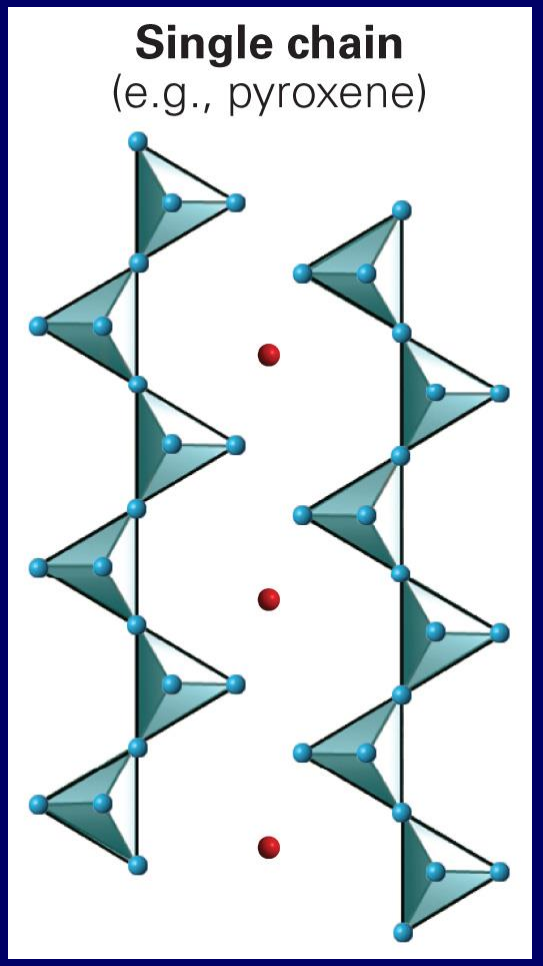
Formation of silicate minerals: Chains



Pyroxene (Diopside)
$CaMgSi_2O_6$



Pyroxene (Orthopyroxene)
$(Mg,Fe)SiO_3$



In single chain silicates, two of the three basal oxygens are bonded together

- Si:O ratio is 1:3

Formation of silicate minerals: double chains



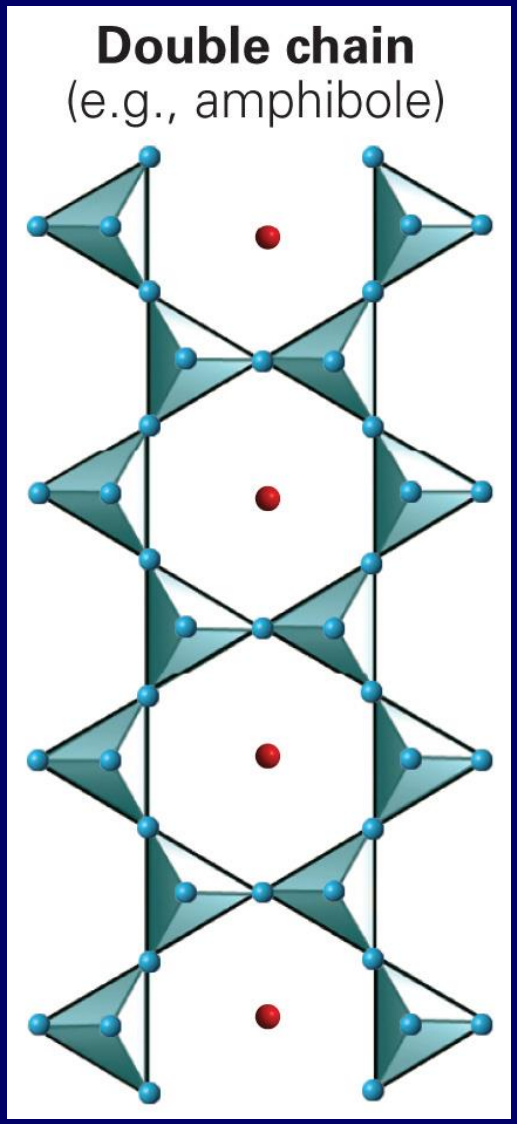
Hornblende, an amphibole

$(Ca,Na)_{2–3}(Mg,Fe,Al)_5(Al,Si)_8O_{22}(OH,F)_2$



Actinolite, an amphibole

$Ca_2(Mg, Fe)_5Si_8O_{22}(OH)_2$



Double-chain silicates are like two single chains that share oxygens where tetrahedra touch to form a double chain

- Si:O ratio is 2:7

Formation of silicate minerals: sheets



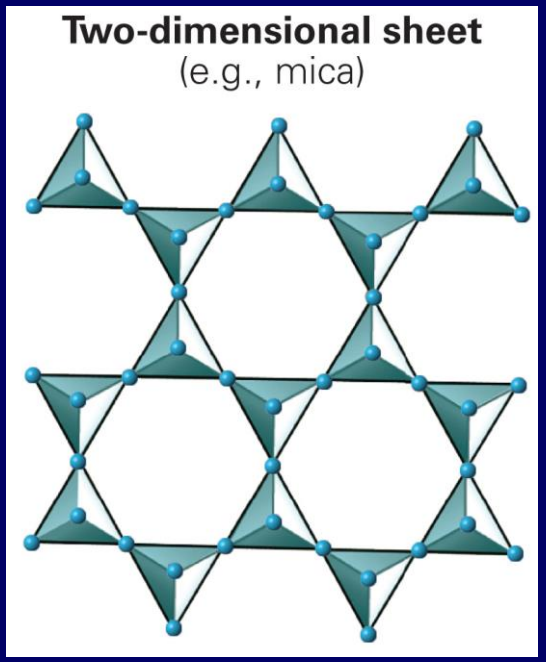
Muscovite, a Mica

$KAl_3Si_3O_{10}(OH,F)_2$



Biotite, a Mica

$K(Mg,Fe)_3AlSi_3O_{10}(OH)_2$



Sheet silicates share 3 oxygens along the base of the tetrahedra, but not the oxygen at the top of the tetrahedra, to form 2-D sheets

- Si:O ratio is 2:5

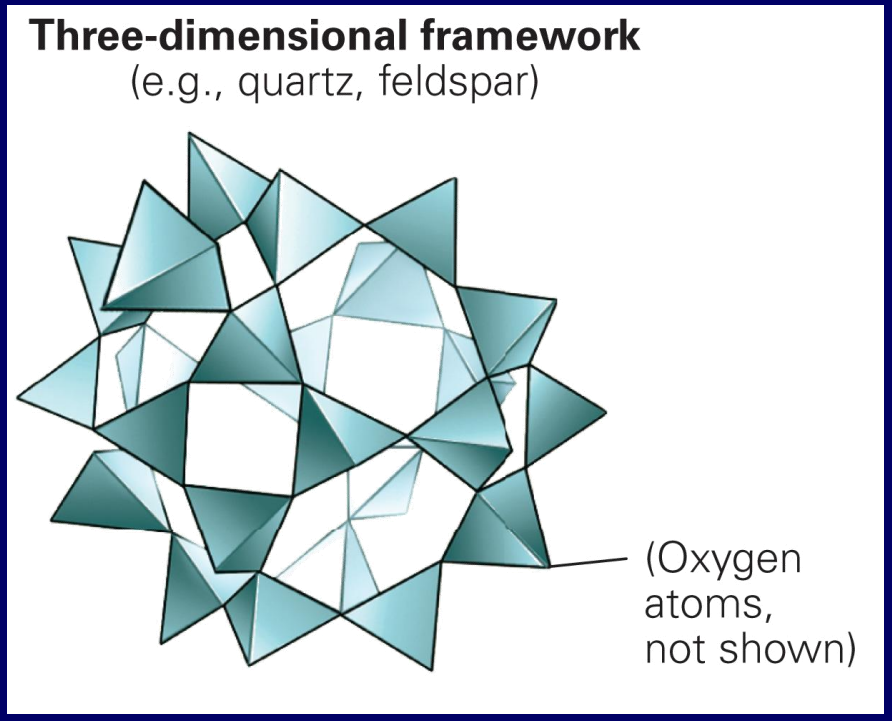


Montmorillonite,
a clay mineral

Formation of silicate minerals: 3D frameworks



Feldspars (Orthoclase)
(KAlSi3O8 – NaAlSi3O8 – CaAl2Si2O8)



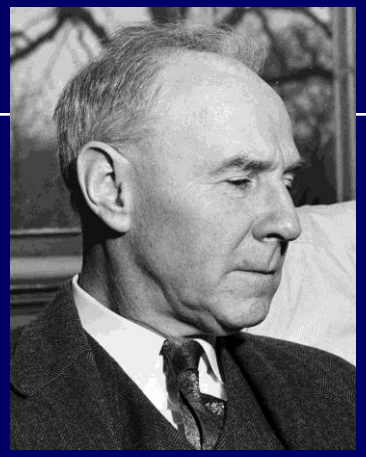
All oxygens are shared between adjacent tetrahedra to form a 3-D framework

- Si:O ratio is 1:2

Quartz: all oxygens are shared by two tetrahedra, no spare oxygens = pure silica

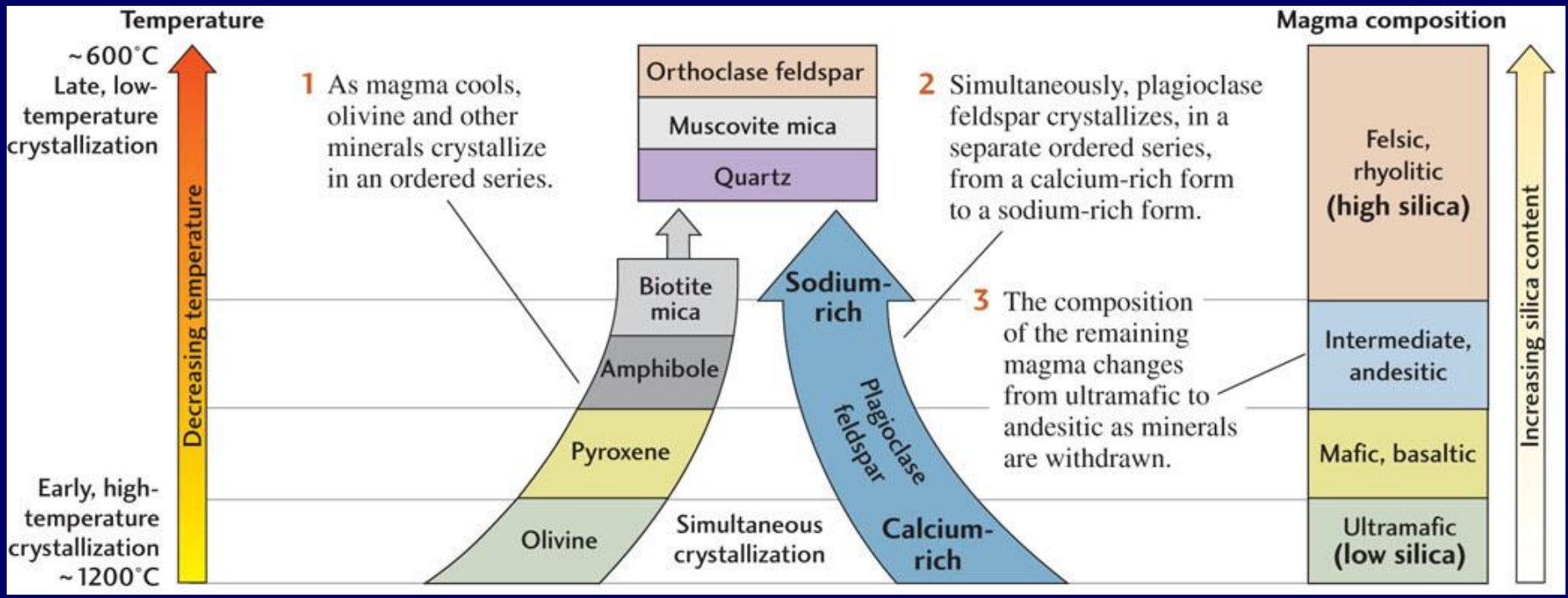


Fractional Crystallisation of Silicate Minerals



Norman Bowen
(1887-1956)

- Minerals crystallise from a melt at different T's
- As they crystallise they change the chemical composition of the remaining melt
- Melt changes composition from 'mafic' to 'felsic'
- Known as a Bowen's reaction series

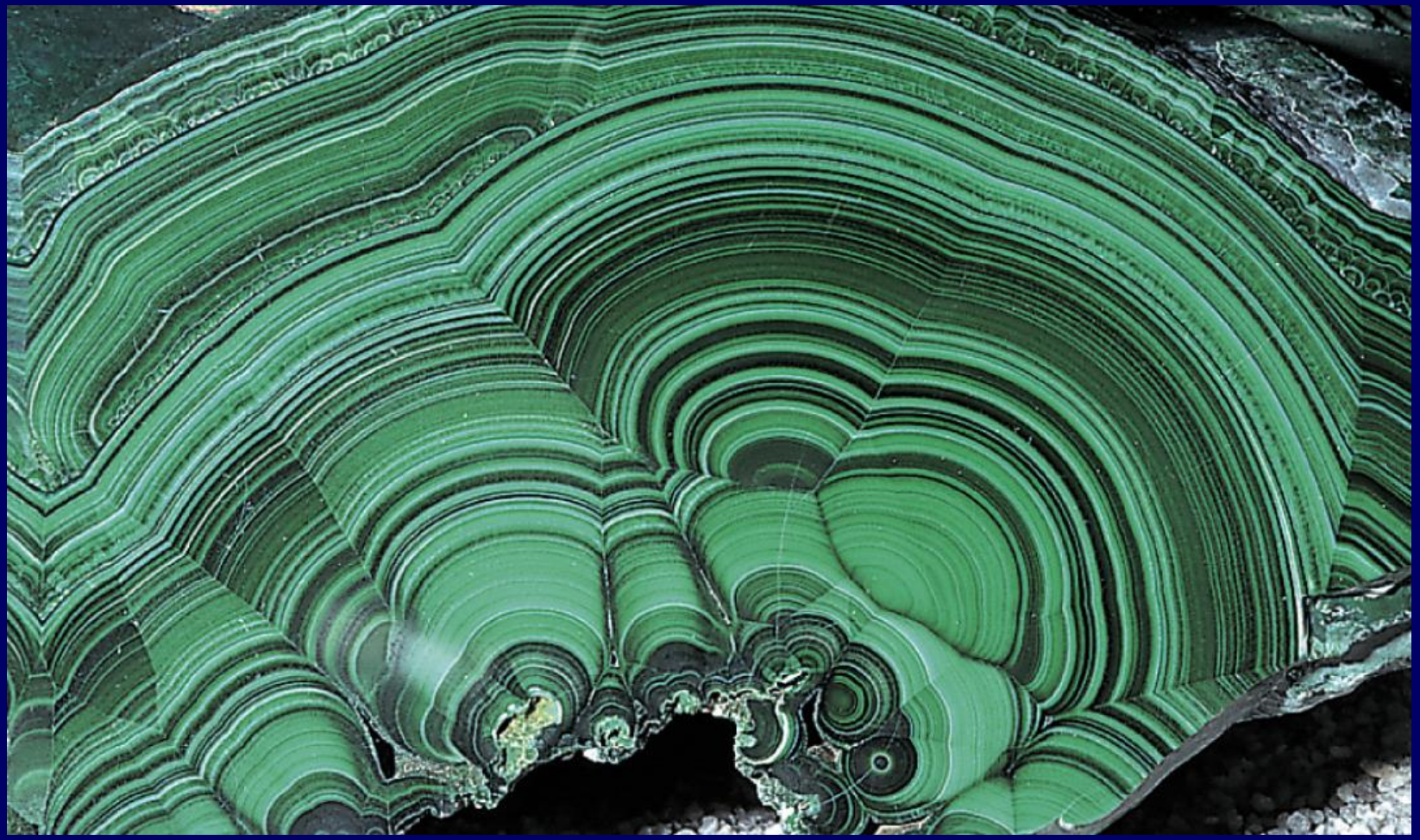


Minerals can be identified from their physical properties

- Colour: determined by chemical composition (ions and trace elements). Visual appearance plus streak. White, clear, colourless
- Lustre: surface appearance (e.g., vitreous, resinous, silky, pearly, metallic, earthy)
- Hardness: strong chemical bonds result in hard minerals. Relative scale (Mohs scale of hardness) 1. Talc to 10. Diamond (quartz = 7)
- Habit: the shape it takes. Depends on planes in crystal structure and typical speed and direction of growth
- Cleavage: planes of weakness in mineral related to atomic structure. Inversely related to bond strength
- Fracture: tendency to break along irregular surfaces other than cleavage planes
- Density: depends on atomic weight of atoms or ions and their closeness of packing in crystal structure
- Other special properties: taste, smell, magnetic, radioactivity

Physical properties reflect their composition and crystal structure

Physical Properties of Minerals: Colour



Malachite is a type of copper ore ($Cu_2[CO_3][OH]_2$); it contains copper plus other chemicals that gives it is diagnostic colour

Physical Properties of Minerals: Colour



Citrine
Yellow



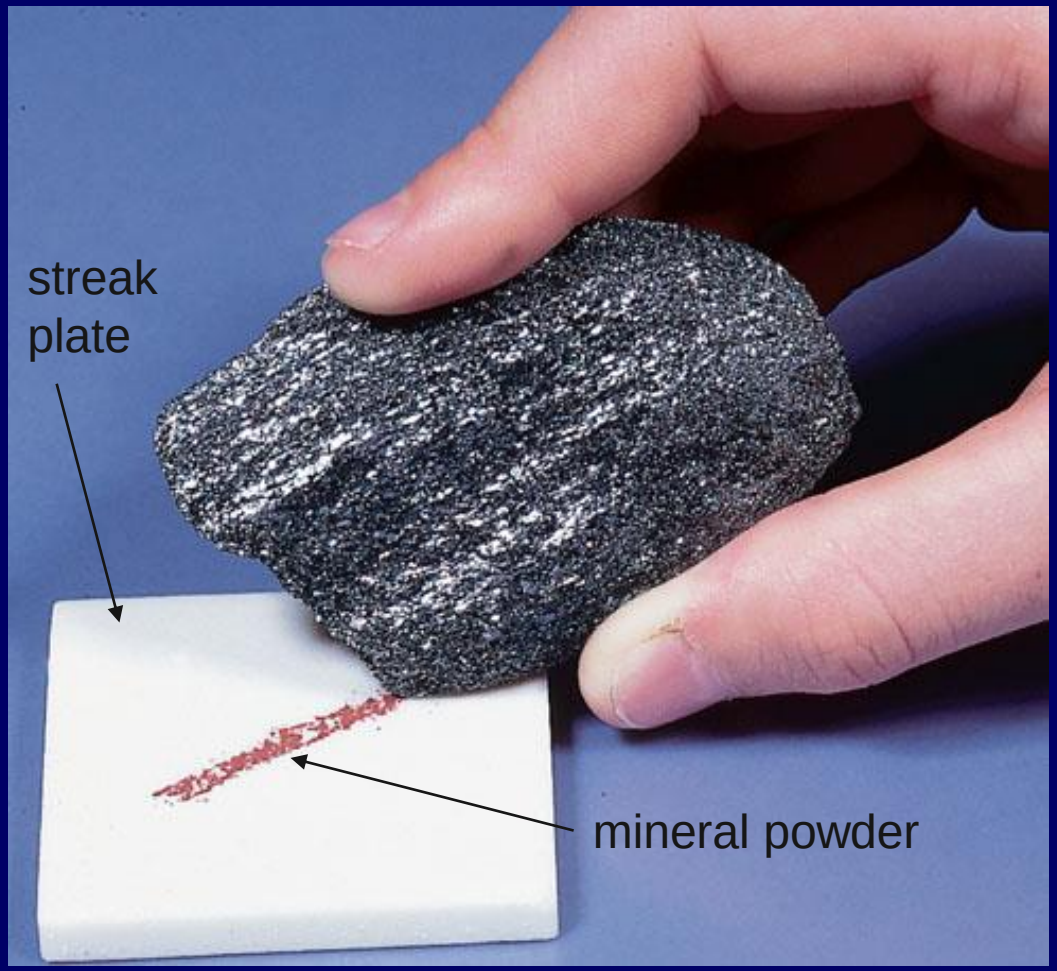
Amethyst
Purple, violet



Smoky quartz
or cairngorm
Brown to grey
opaque

Variation in the colour of quartz due to impurities

Physical Properties of Minerals: Streak



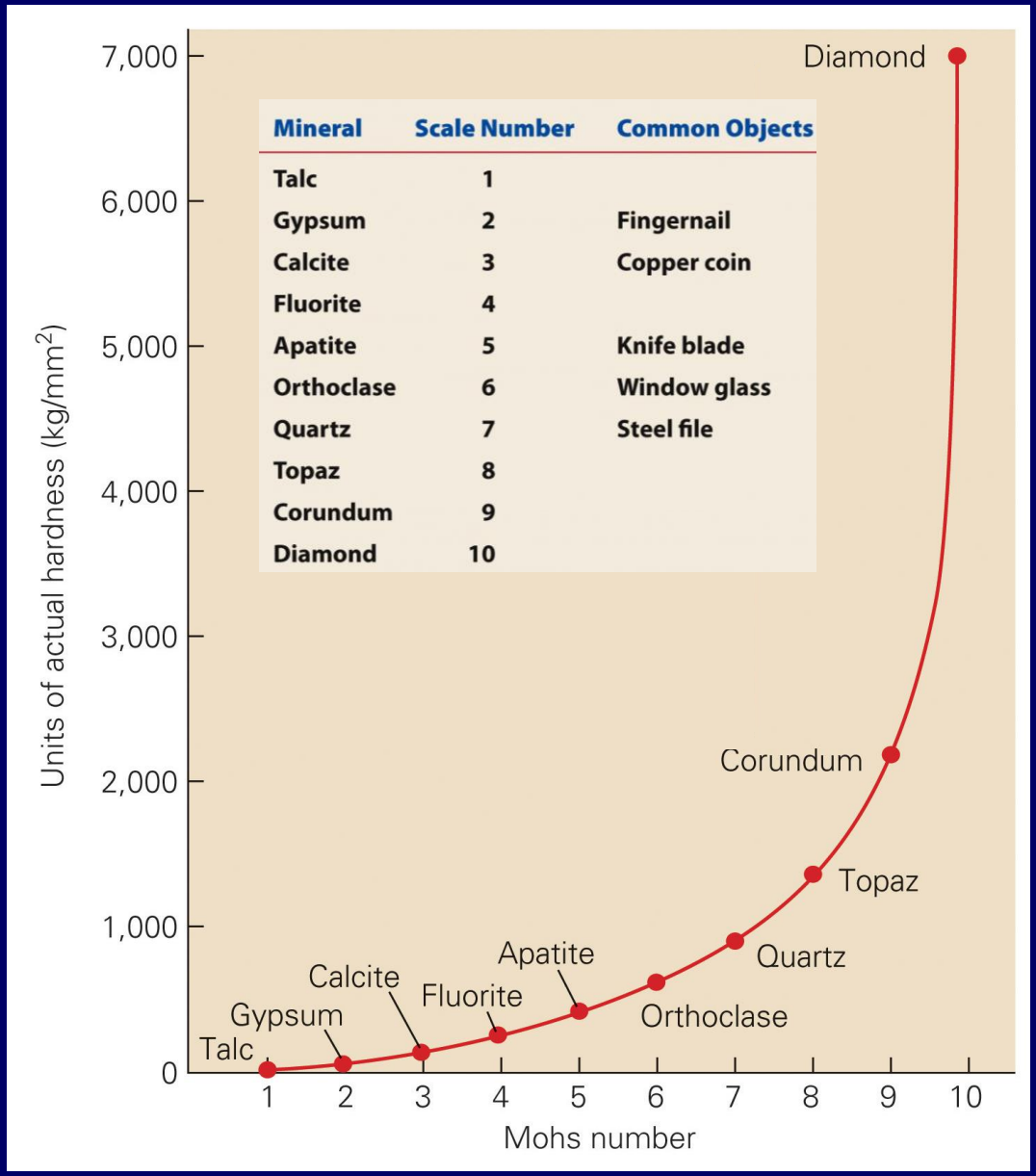


Hematite may look black, red or brown but always leaves a reddish brown streak

Physical Properties of Minerals: Hardness



Friedrich Mohs
(1773-1839)



Mohs hardness scale

Physical Properties of Minerals: Crystal Habit



Physical Properties of Minerals: Cleavage



Cleavage varies inversely with bond strength

(Ionic bonds weak = good cleavage; covalent bonds strong = poor or no cleavage

Physical Properties of Minerals: Other properties

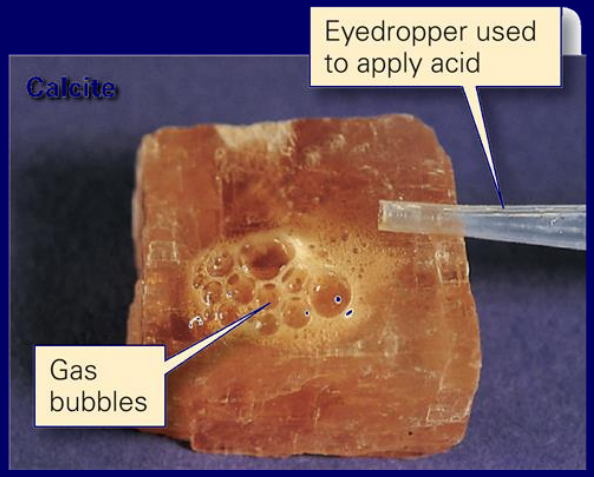


Taste
e.g., Halite – ‘rock salt’
(Halide mineral)
NaCl

Magnetic
e.g., Magnetite
(Oxide mineral)
Fe_3O_4



The magnetism attracts nails



Effervescence (reactivity with acid)
e.g., Calcite (Carbonate mineral) $CaCO_3$



Radioactive
e.g., Autunite (Phosphate mineral)
$Ca(UO_2)_2(PO_4)_2 \cdot 10–12H_2O$

Minerals as natural resources



Summary: Minerals

- Minerals are the building blocks of rocks
- Defined as a naturally occurring, solid, crystalline substance, usually inorganic, with a specific chemical composition
- Minerals are classified according to their chemical composition into one of 8 groups
- The silicates group are quantitatively the most important group within the Earth's crust
- Different minerals melt and crystallize at different temperatures
- As crystals form, they extract certain elements from the melt
- The chemical composition of the melt progressively changes from mafic (low silica) to felsic (high silica)
- The physical properties of minerals are due to their composition and crystal structure, and can be used to identify them

Intended Learning Outcome Revision

Now we have completed the class, you should be able to:

- explain the physical basis for the definition of a mineral and use this to distinguish minerals from non-minerals
- outline the different ways in which minerals form
- name the five most common classes of rock-forming minerals and give an example of a mineral that belongs to that class
- describe the crystallisation of silicate minerals from a melt
- name the main physical properties of minerals and give examples to explain how these could be used to distinguish minerals

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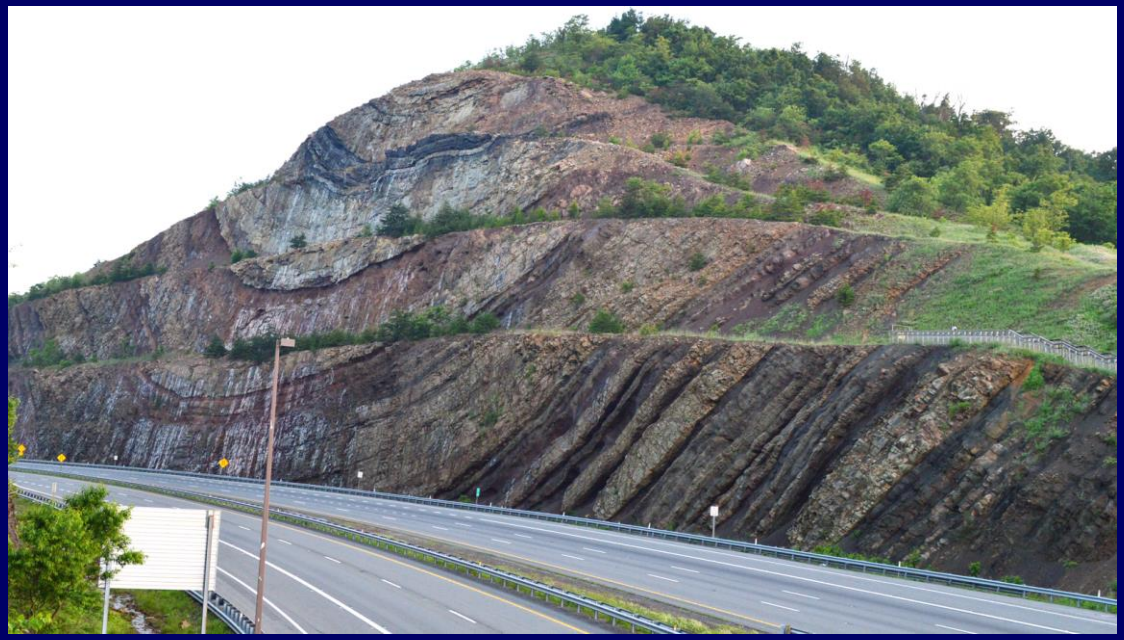
How to Build a Planet

Week 4 An Introduction to Rocks

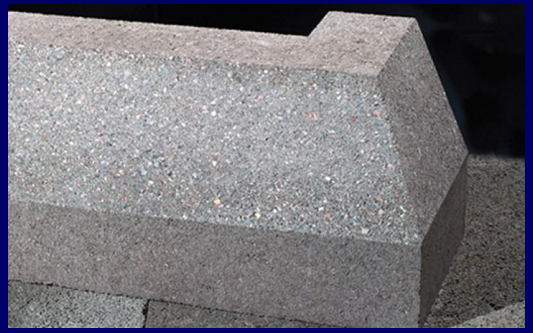
- physical basis for the definition of a rock
- how you can identify different types of rocks on the basis of their texture and composition
- classify rocks according to their origin,
- discuss the geologic environments in which rocks form



A rock is naturally occurring











A rock is naturally occurring











A rock is solid



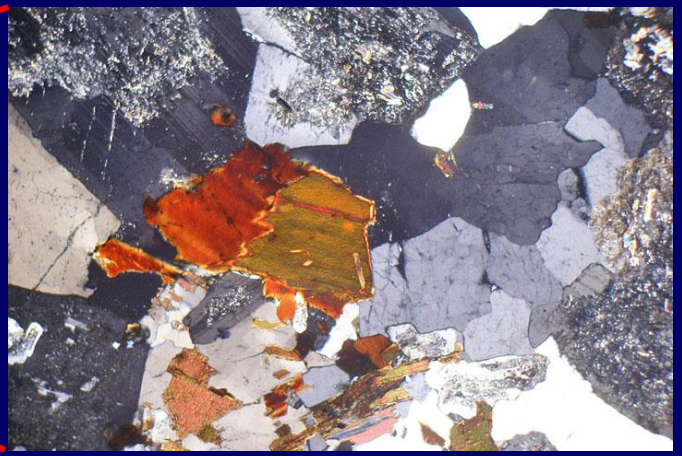


A pile of loose mineral grains (sediment) is not a rock

A rock is solid: crystalline



Skiddaw Granite, Cumbria



A photomicrograph shows the rock is held together by interlocking crystals

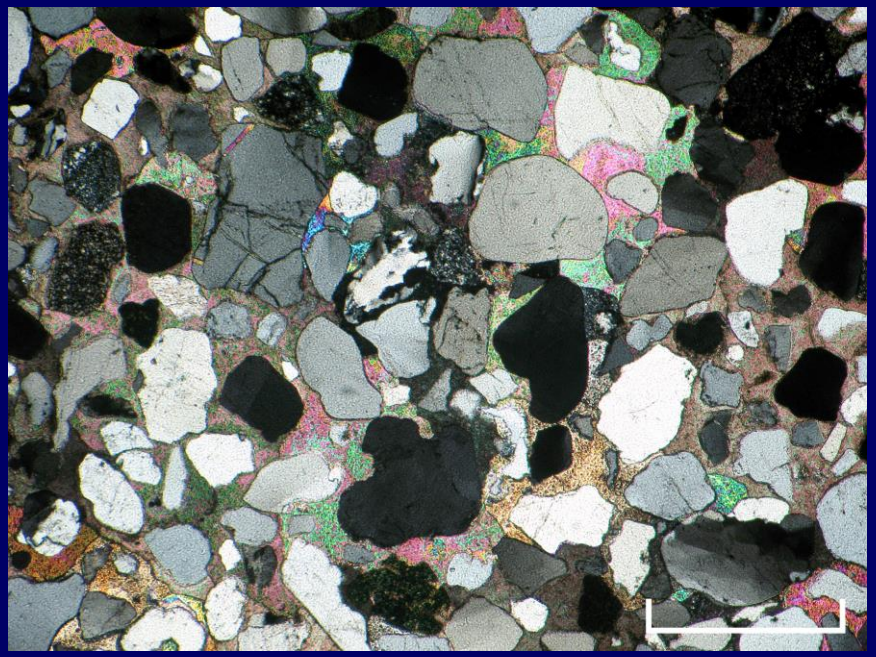
A rock is solid: cemented

Loose sediment





Quartz sandstone



held together by calcite cement between mineral grains

Lithification: the process of turning sediment into rock

What are rocks?

A rock is an aggregate of mineral grains, or in some cases, non-mineral solid matter (glass, fossils)



Rock (Granite)

Chemical composition of rocks in the Earth

Element	% by weight	
	Mantle	Crust (continental)
O	44.8	46.6
Si	21.5	27.7
Mg	22.8	1.5
Fe	5.8	5.0
Al	2.2	8.1
Ca	2.3	3.6
Na	0.3	2.8
K	0.03	2.6
Other	0.3	2.1

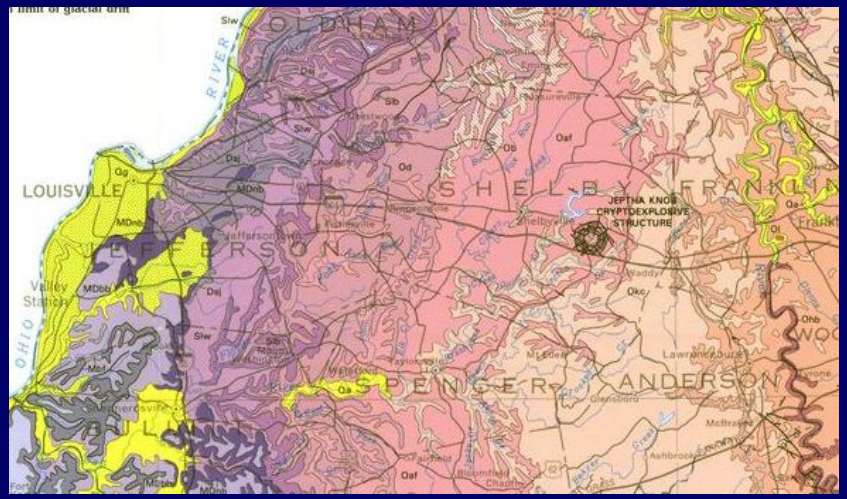
Terminology to describe rocks at the Earth's surface



Natural exposure of bedrock in a river



Manmade exposure of bedrock in a quarry face



Outcrop pattern of on a map



Rocks crop out and are exposed

Rocks are Classified by their Origin



Igneous rocks



Sedimentary rocks



Metamorphic rocks

Igneous rocks- solidification from molten rock



Sedimentary rocks- erosion & deposition of pre-existing rocks



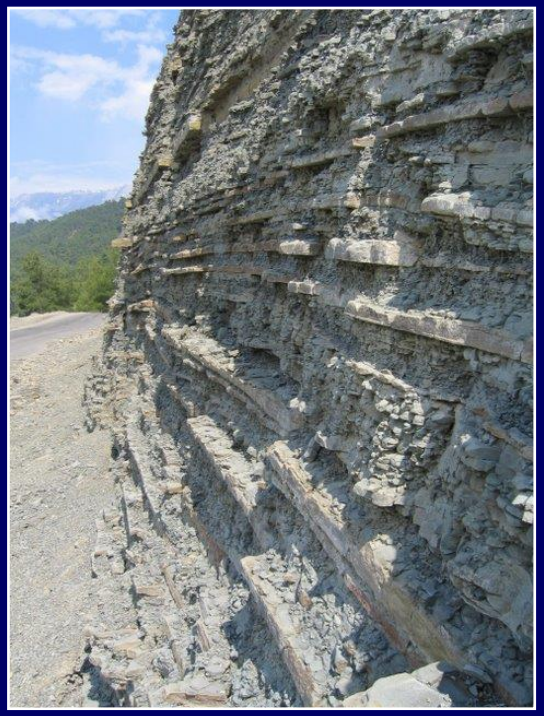
Metamorphic rocks- changes due to heat and/or pressure



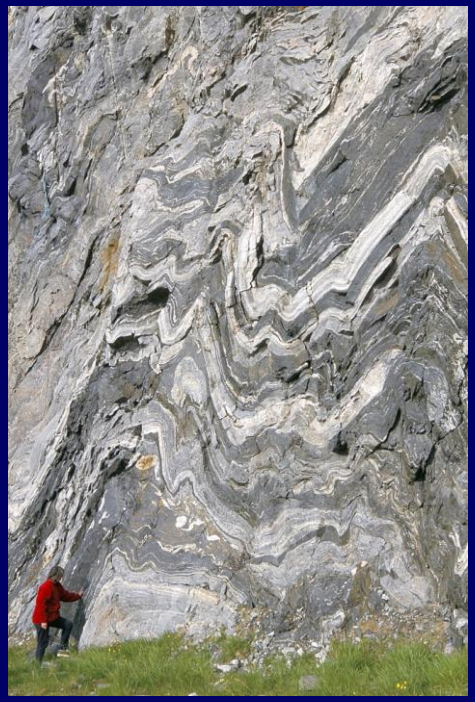
Identifying the different rock types by texture



Igneous- Crystalline



Sedimentary
Non-crystalline and layered
Layering: due to grain size



Metamorphic
Crystalline and layered
Layering: due to composition

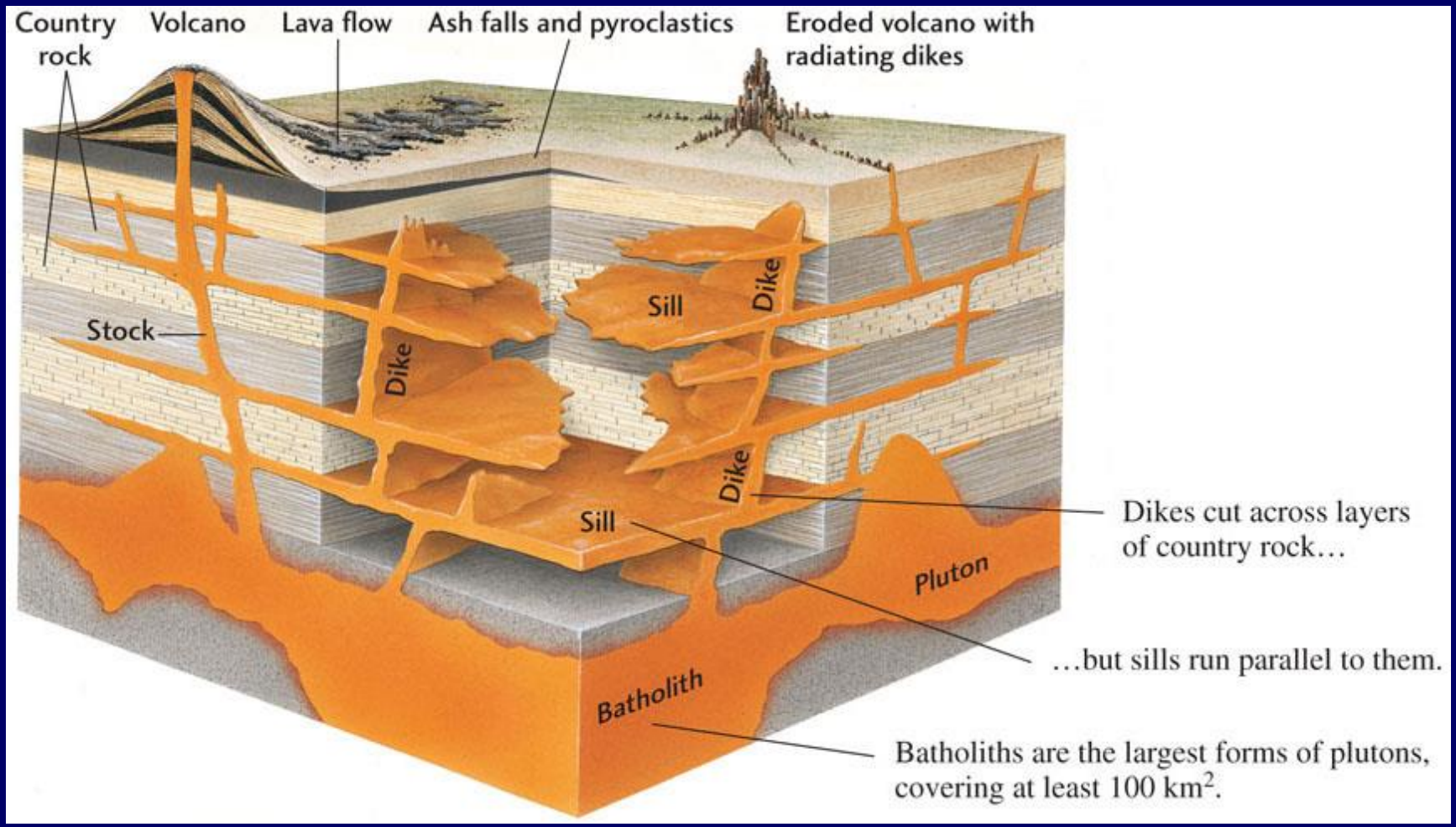
Identifying the different rock types by composition

Igneous Rocks	Sedimentary Rocks	Metamorphic Rocks
Quartz	Quartz	Quartz
Feldspar	Clay minerals	Feldspar
Mica	Feldspar	Mica
Pyroxene	* Calcite	Garnet
Amphibole	* Dolomite	Pyroxene
Olivine	* Gypsum	Staurolite
	* Halite	Kyanite

Note: **Asterisks indicate nonsilicate minerals.**

Some common minerals of Igneous, sedimentary and metamorphic rocks

About Igneous Rocks

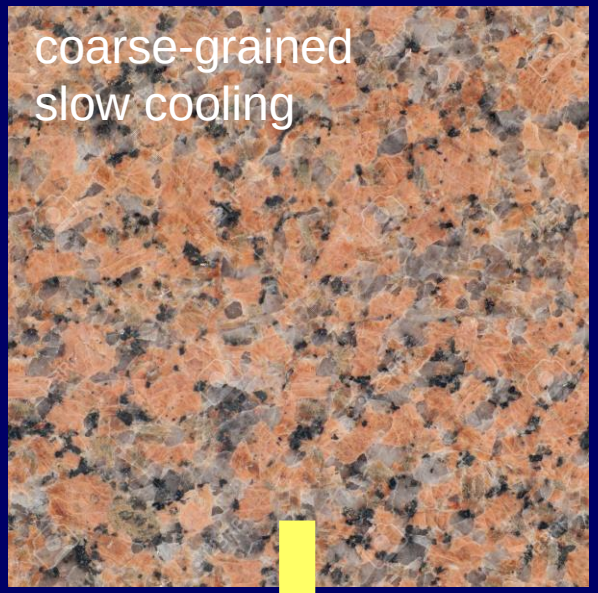


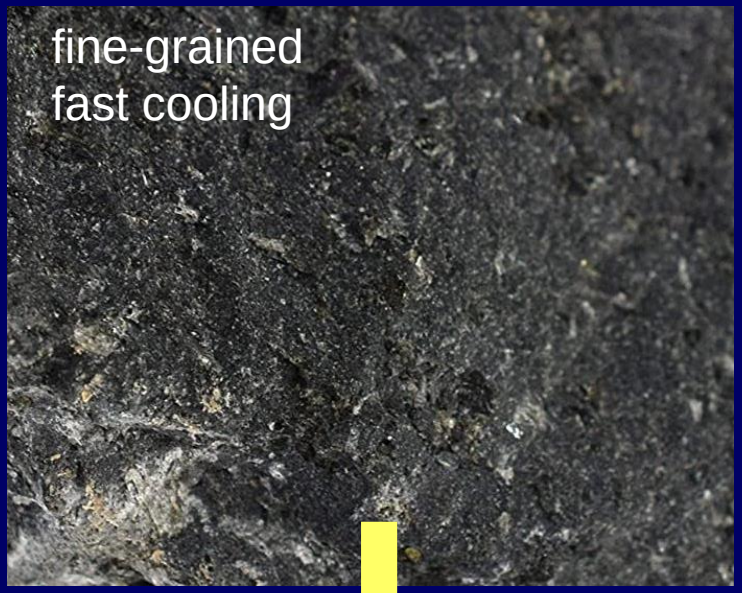
Igneous rock types: texture

Granite

Basalt

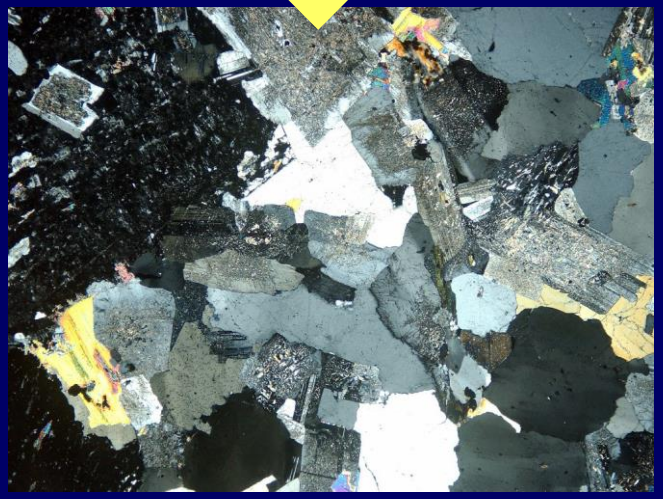
Seen with a magnifying glass

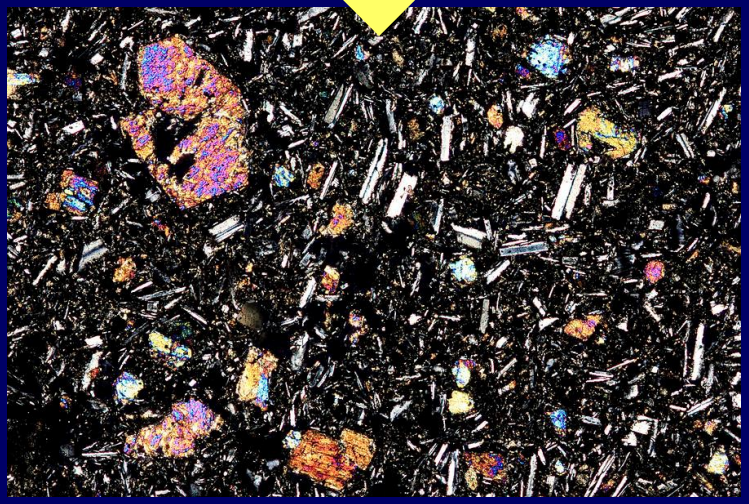




1 cm

Seen with a polarizing microscope





1 mm

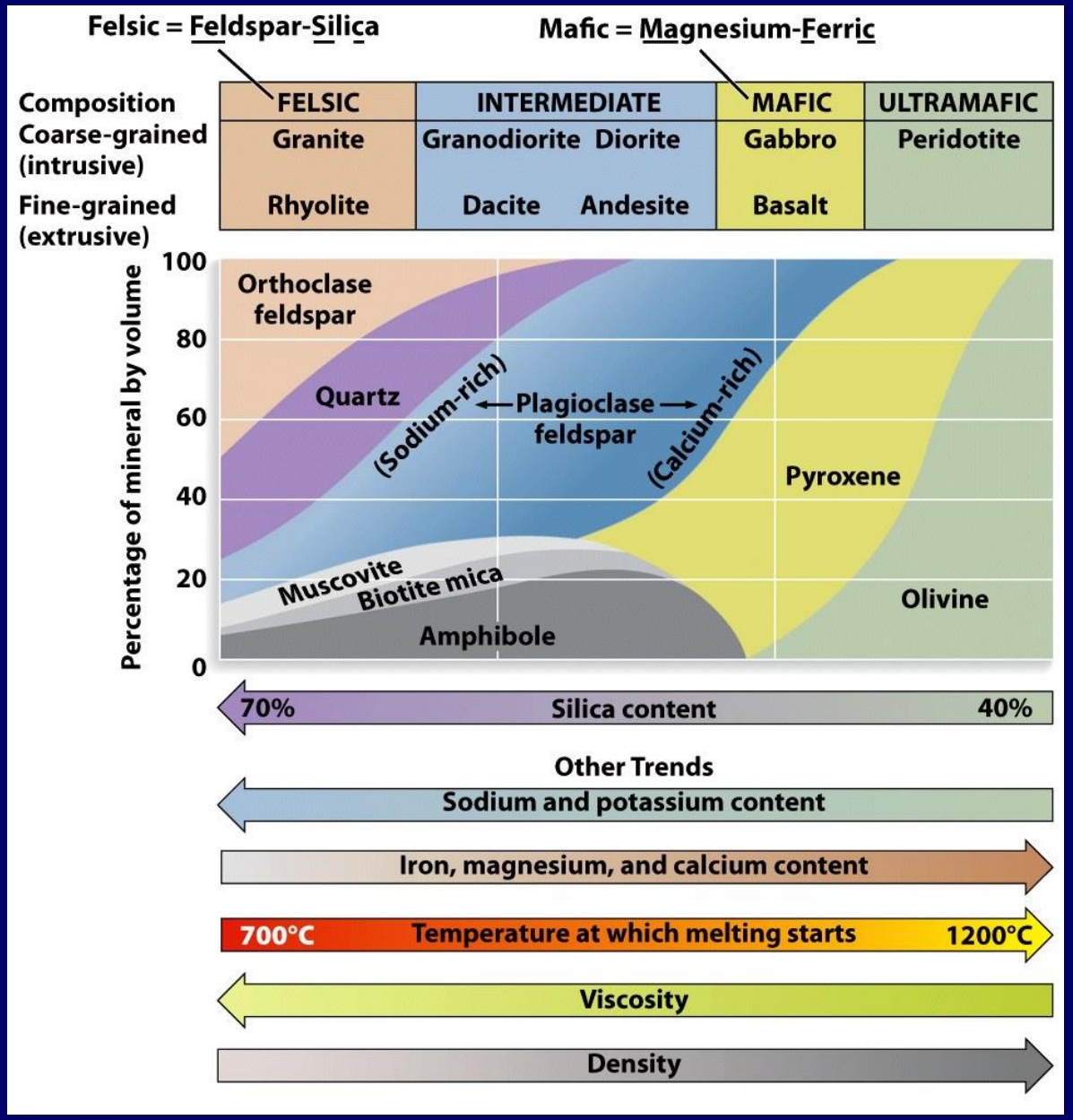
Igneous rock types: chemical & mineralogical composition

Felsic

- Rich in silica, low in Mg + Fe
- Quartz
- Feldspar
- Muscovite

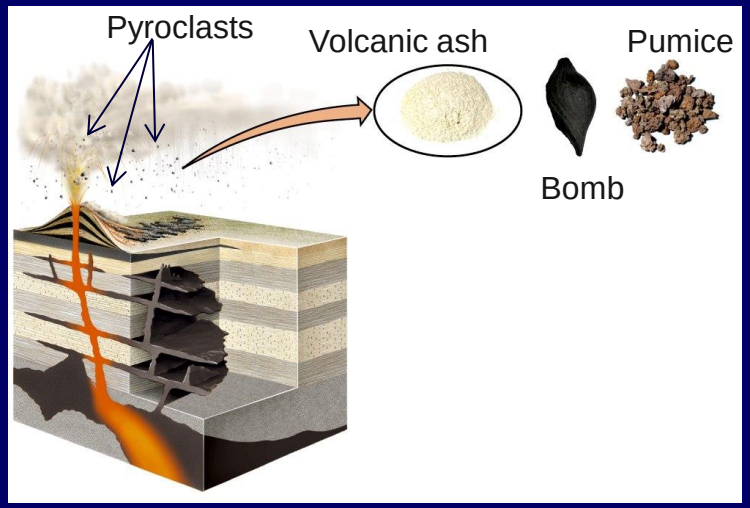
Mafic

- Poor in silica, rich in Mg + Fe
- Biotite
- Amphibole group
- Pyroxene group
- Olivine



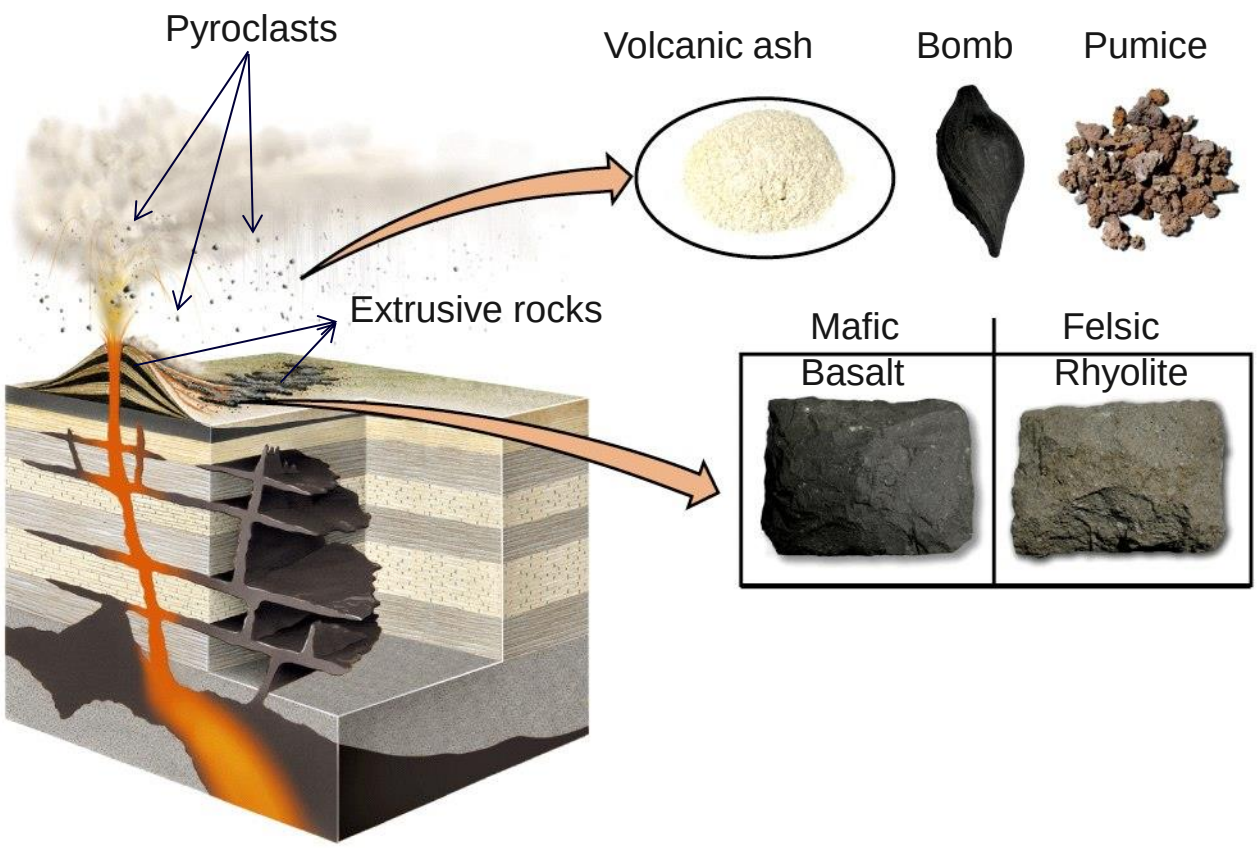
Extrusive pyroclasts: igneous or sedimentary?





Mount Saint Helens,
USA 1980

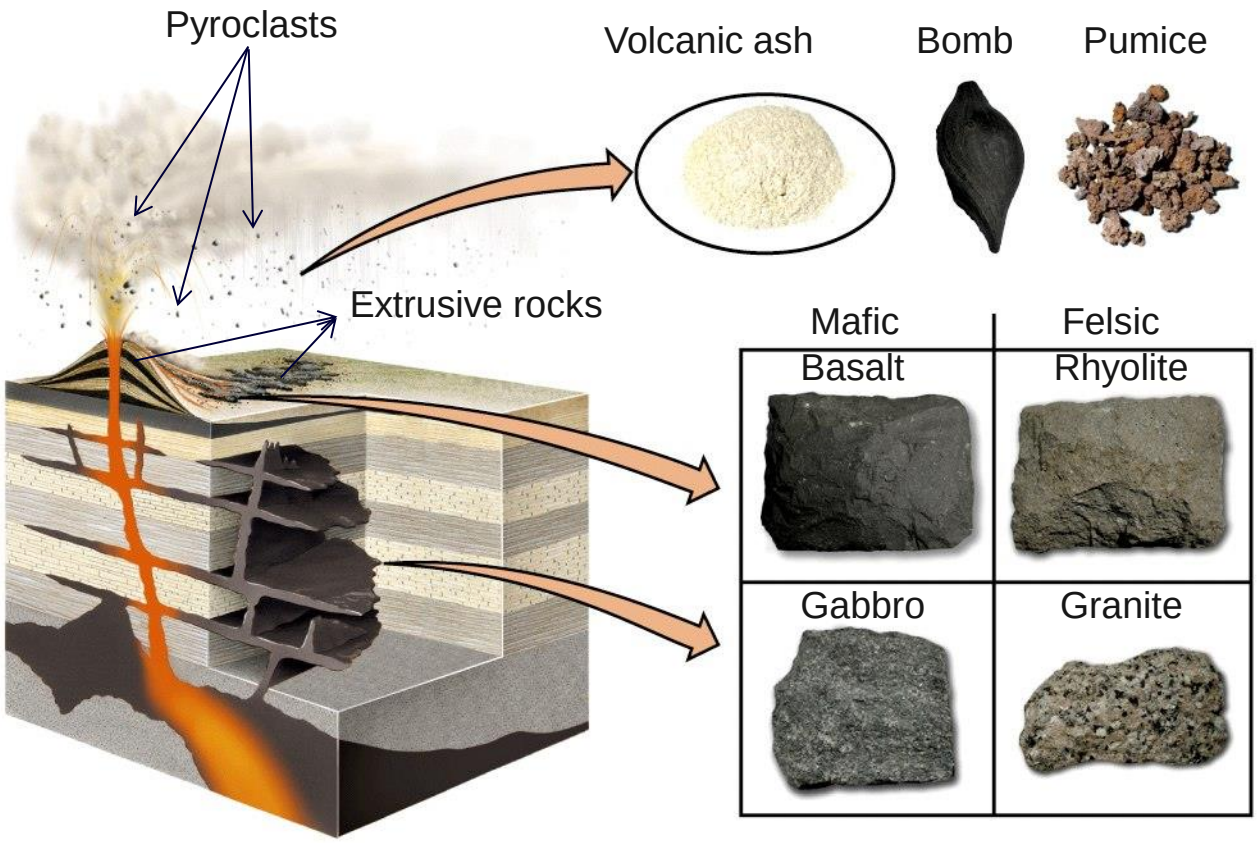
Extrusive igneous rocks



Extrusive pyroclasts form in violent eruptions from lava in the air

Extrusive igneous rocks cool rapidly and are fine-grained

Intrusive igneous rocks

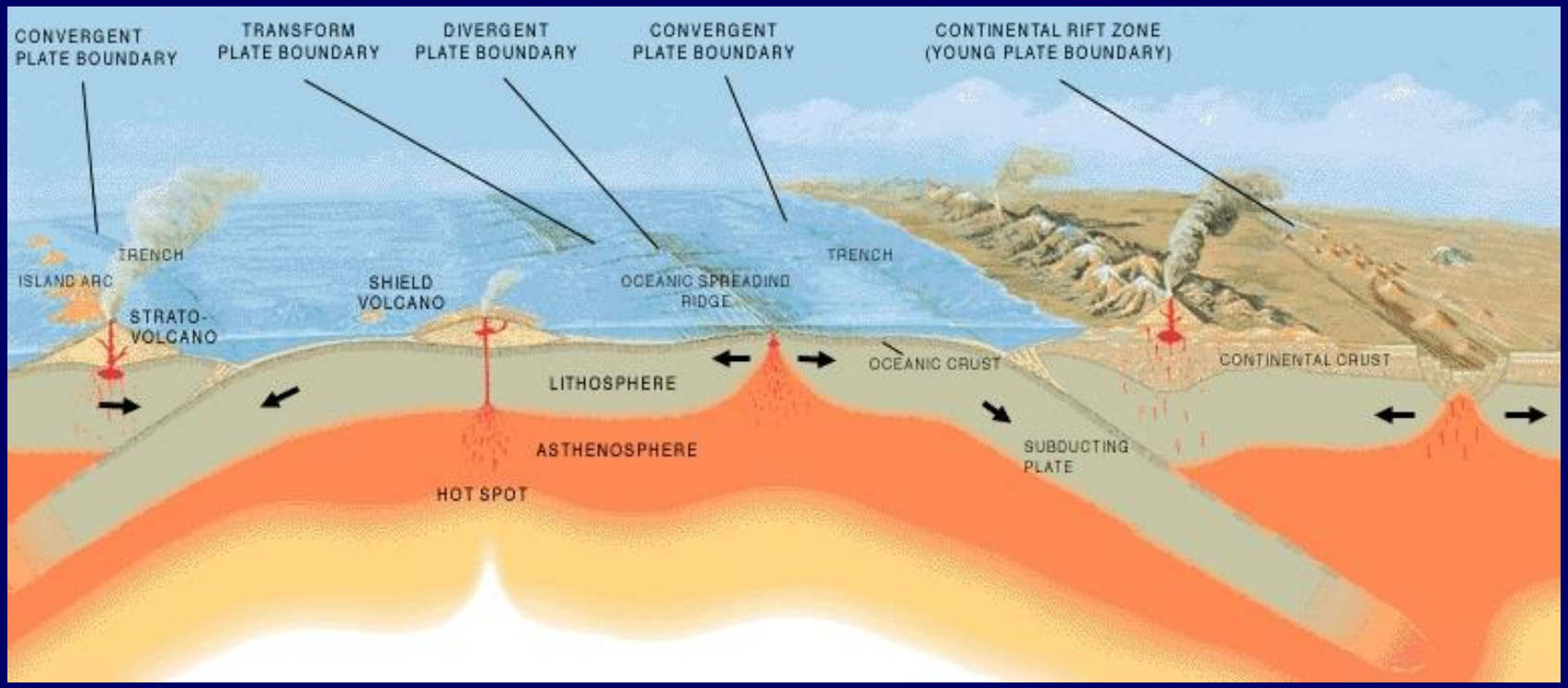


Extrusive pyroclasts form in violent eruptions from lava in the air

Extrusive igneous rocks cool rapidly and are fine-grained

Intrusive rocks

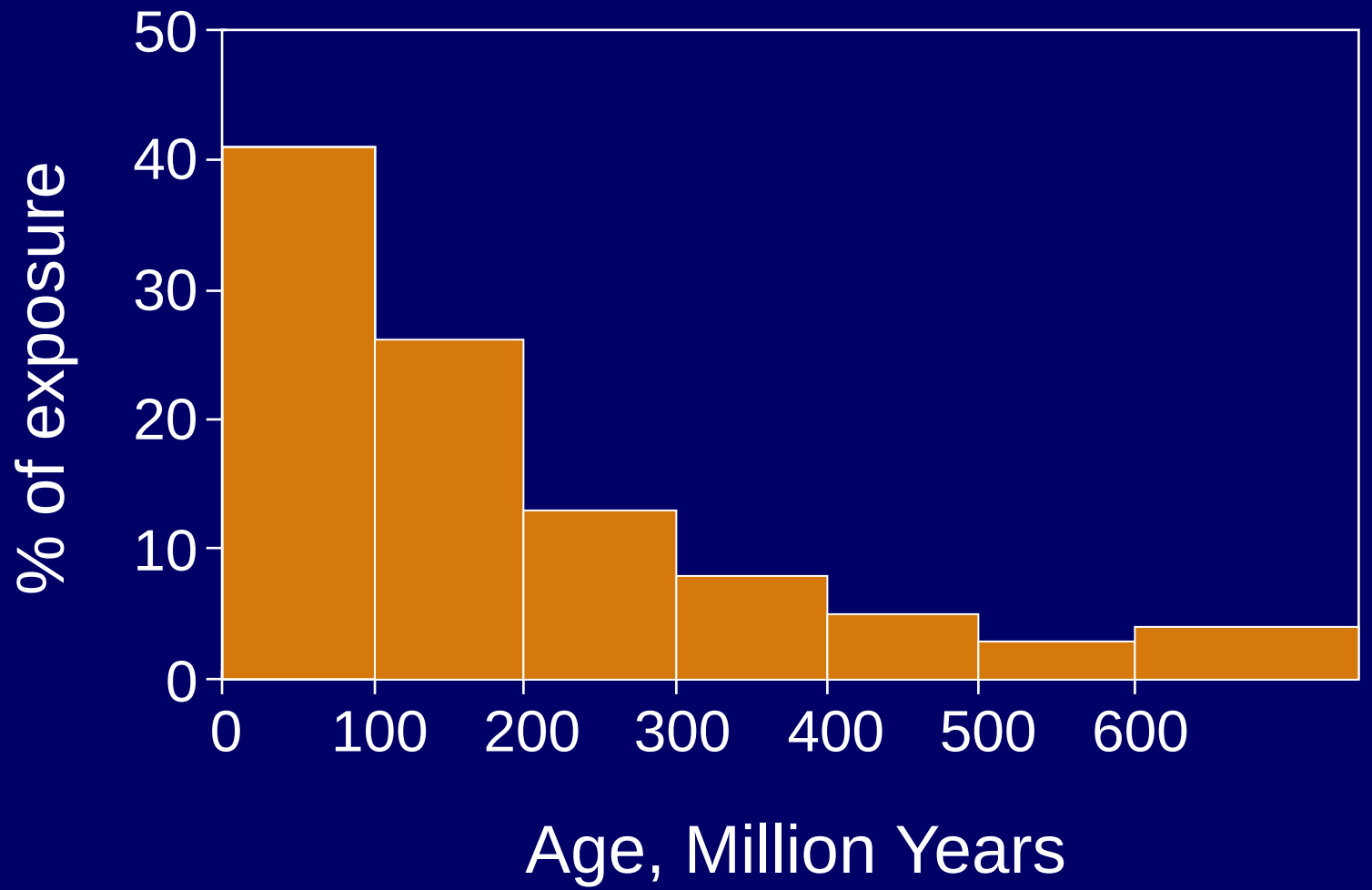
Main settings of igneous activity





Pillow lava erupted onto the ocean floor

Sedimentary Rocks: Crustal Abundance



Sedimentary rocks:

- 5% of the Earth's crust
- cover about 75% of the Earth's surface (66% of continental areas)

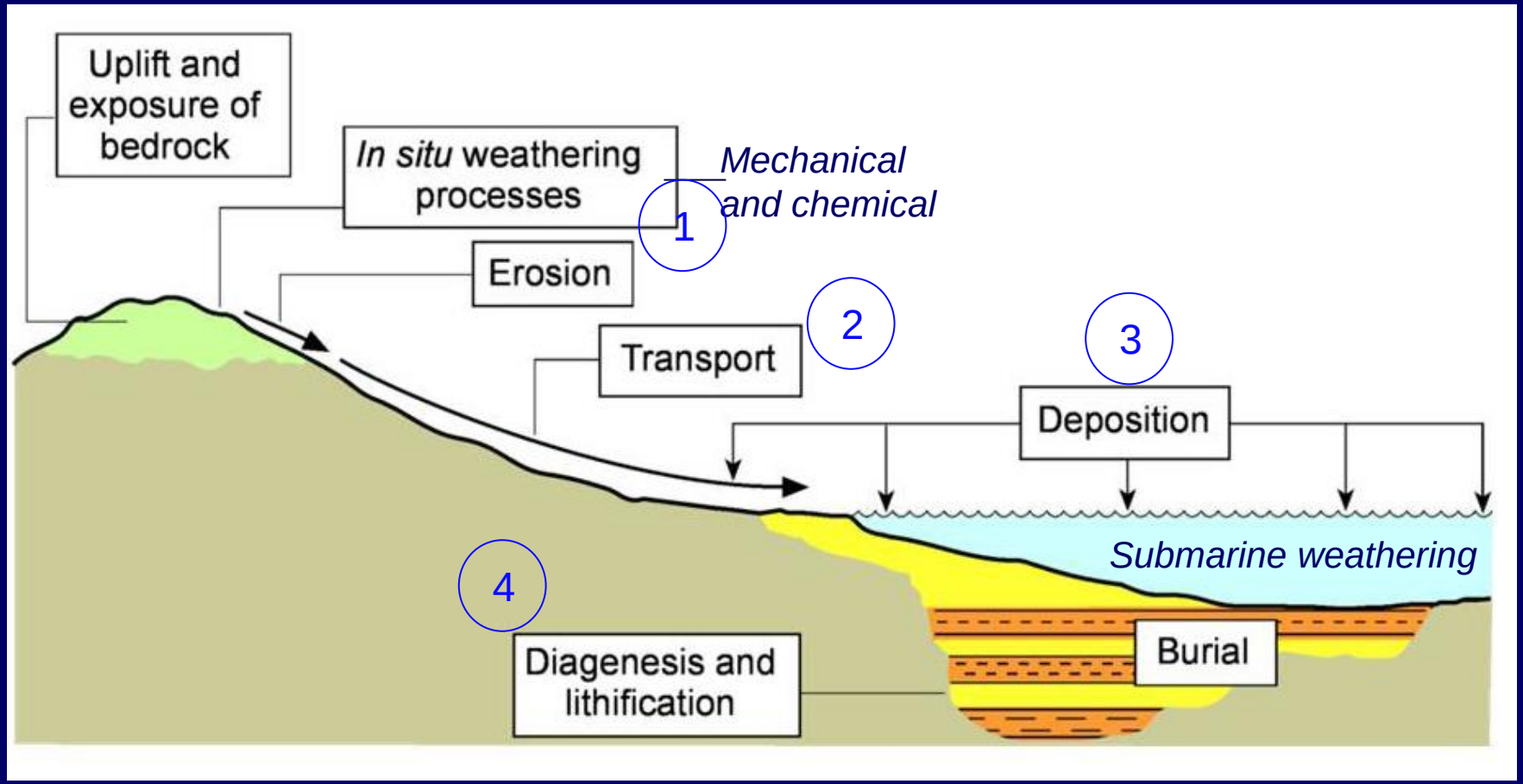
Main categories of sedimentary rocks

Clastic = consist of fragments of pre-existing rocks, transported and deposited by physical processes e.g. sandstones

Biogenic: remains of living organisms e.g. limestone, chalk, coal, oil

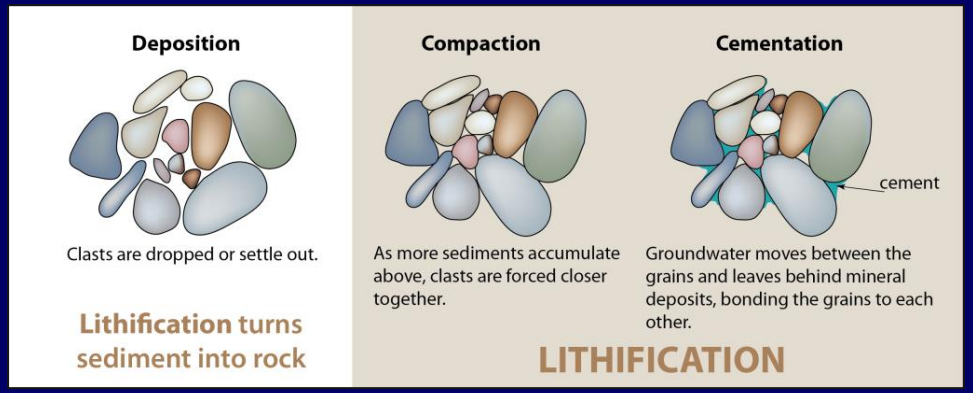
Chemical: precipitate directly out of water e.g. evaporites

Weathering and Water-Rock Interactions



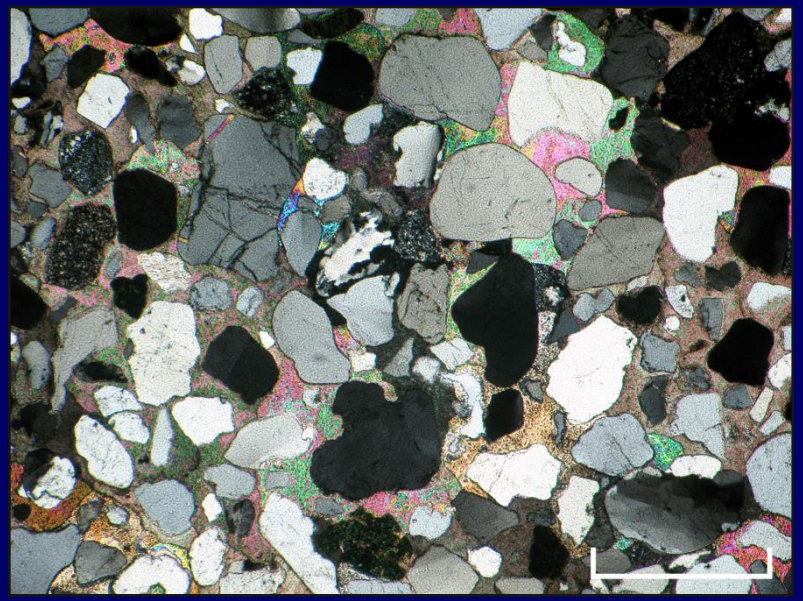
Turning sediment to rocks: diagenesis and lithification

- **Lithification -** conversion of sediment to rock
- **Diagenesis** – processes occurring during lithification









Cementation

Clastic sedimentary rocks: roundness, sorting, size



Coarse-grained clastic rock made up of poorly sorted, rounded rock fragments

- Clastic rocks are made-up dominantly of material transported (wind, water, ice) as discrete particles of rock or mineral
- Roundness, sorting (similarity) and grain size are indicators of distance transported and energy of transport



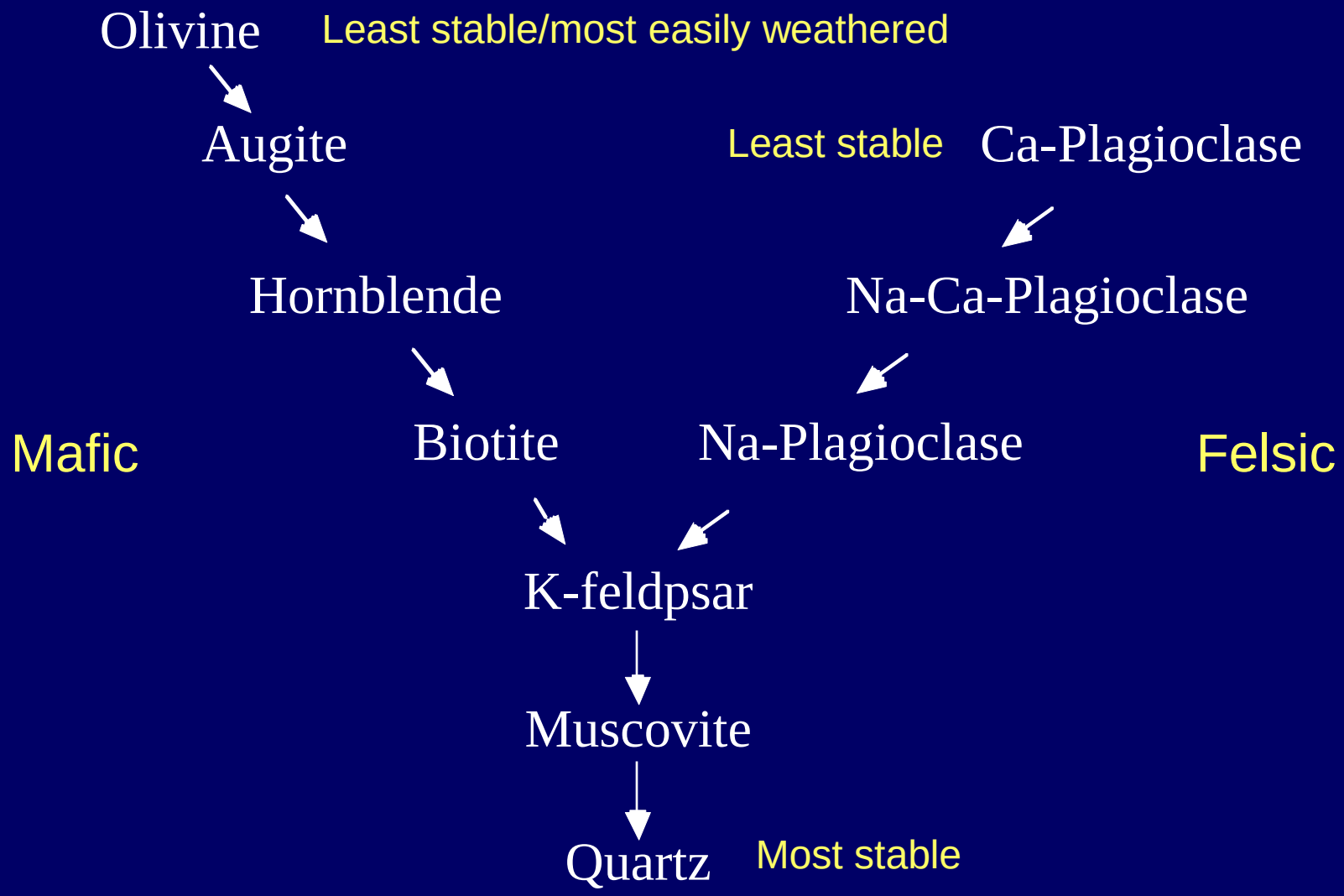
Fine-grained terrigenous clastic rock made up of layers of well-sorted mudstones and sandstones

Effects of weathering on mineralogy

Mineralogy	Igneous and Metamorphic	Sandstone
Feldspar	60%	15%
Quartz	25%	50%
Ca,Mg/Fe silicates	15%	0%

Differences means not all minerals weather equally

Weathering series



Clastic sedimentary rocks: grain size

sediment

rock









mudstone (shale)



sandstone



conglomerate

low

high

Energy of flow at time of deposition

Chemical sedimentary rocks: evaporites



Evaporites formed due to evaporation of a saline lake



Evaporites interbedded with thin dark mudstones

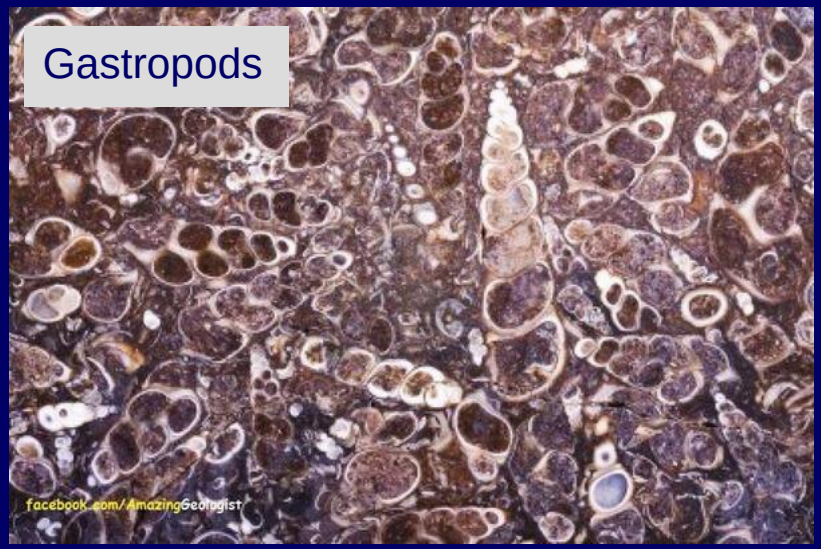
Biogenic sedimentary rocks: limestones

- Limestone is a non-clastic rock formed by biochemical precipitation of calcite ($CaCO_3$) from organisms (shells), or by direct chemical precipitation from marine and other waters saturated with calcium carbonate

- Chalk: is made up of Foraminefera (single-cell organisms) and is very fine grained

- Fossiliferous limestone: is formed by the cementation of shells and is medium to coarse grained

Biogenic limestones

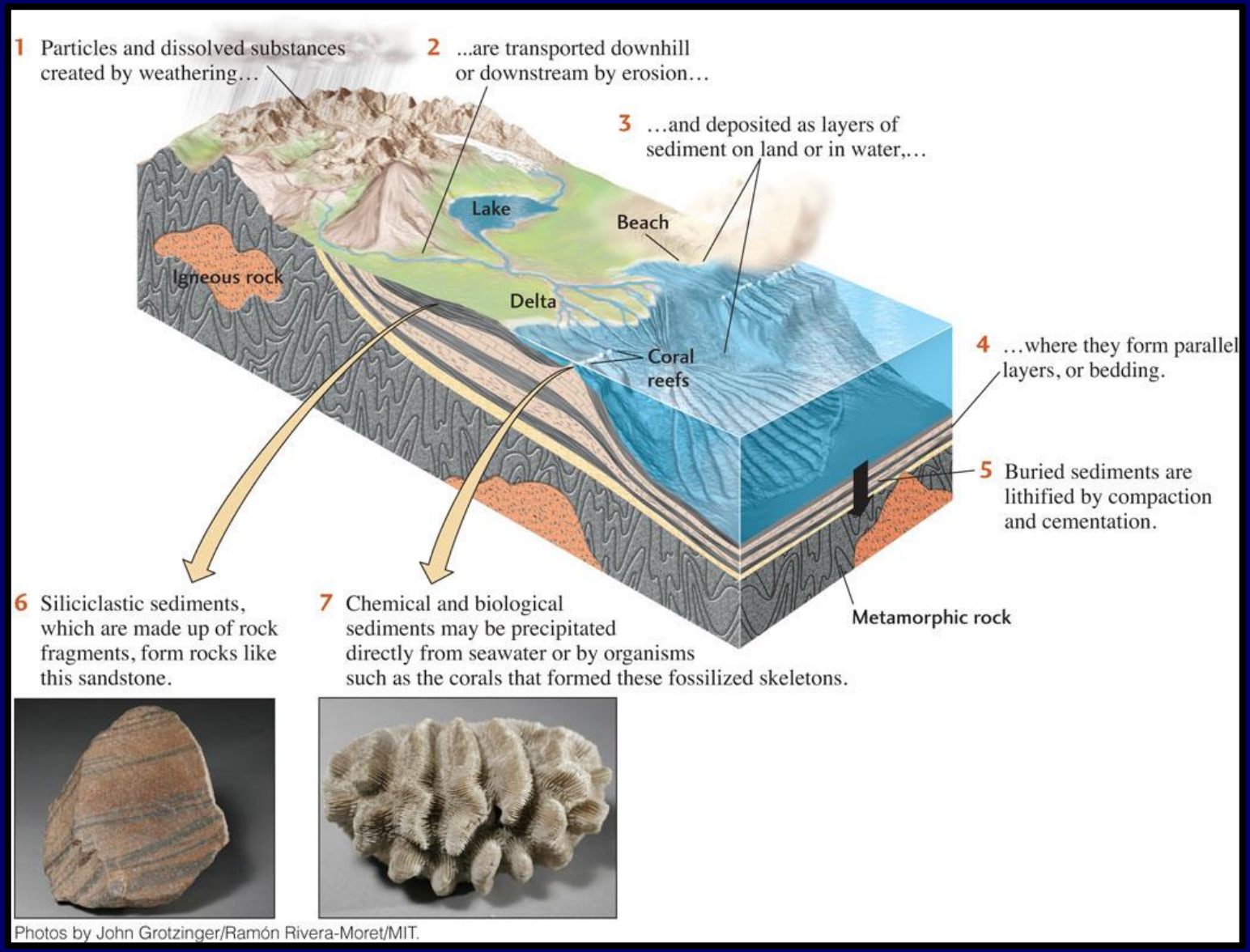








Places where sedimentary rocks are deposited



Photos by John Grotzinger/Ramón Rivera-Moret/MIT.

What are metamorphic rocks?



Metamorphic rocks – changed by the action of heat and/or pressure

Metamorphism can cause minerals to recrystallize



Limestone

undergoes

metamorphism



Marble



Sandstone

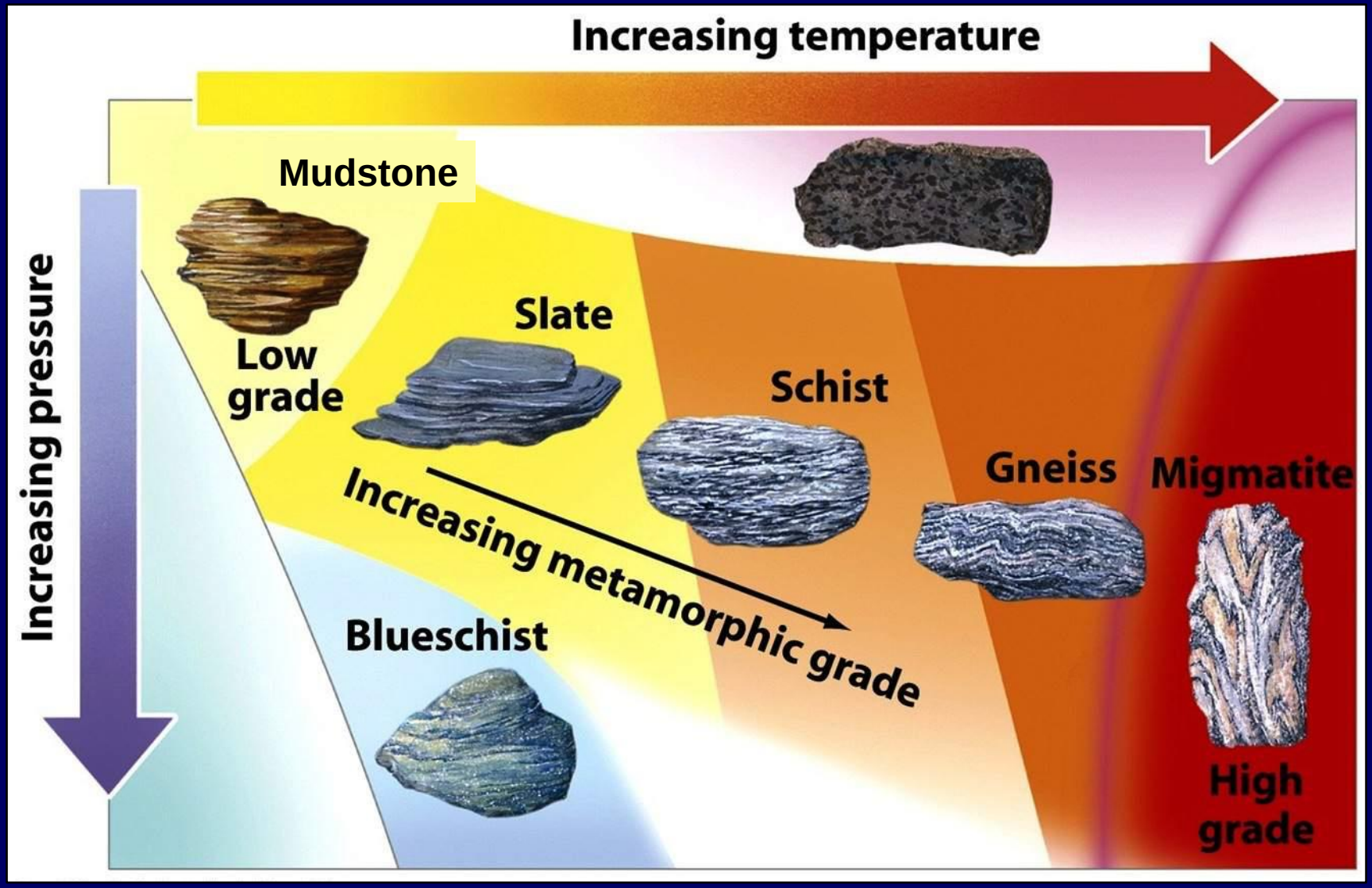
undergoes

metamorphism

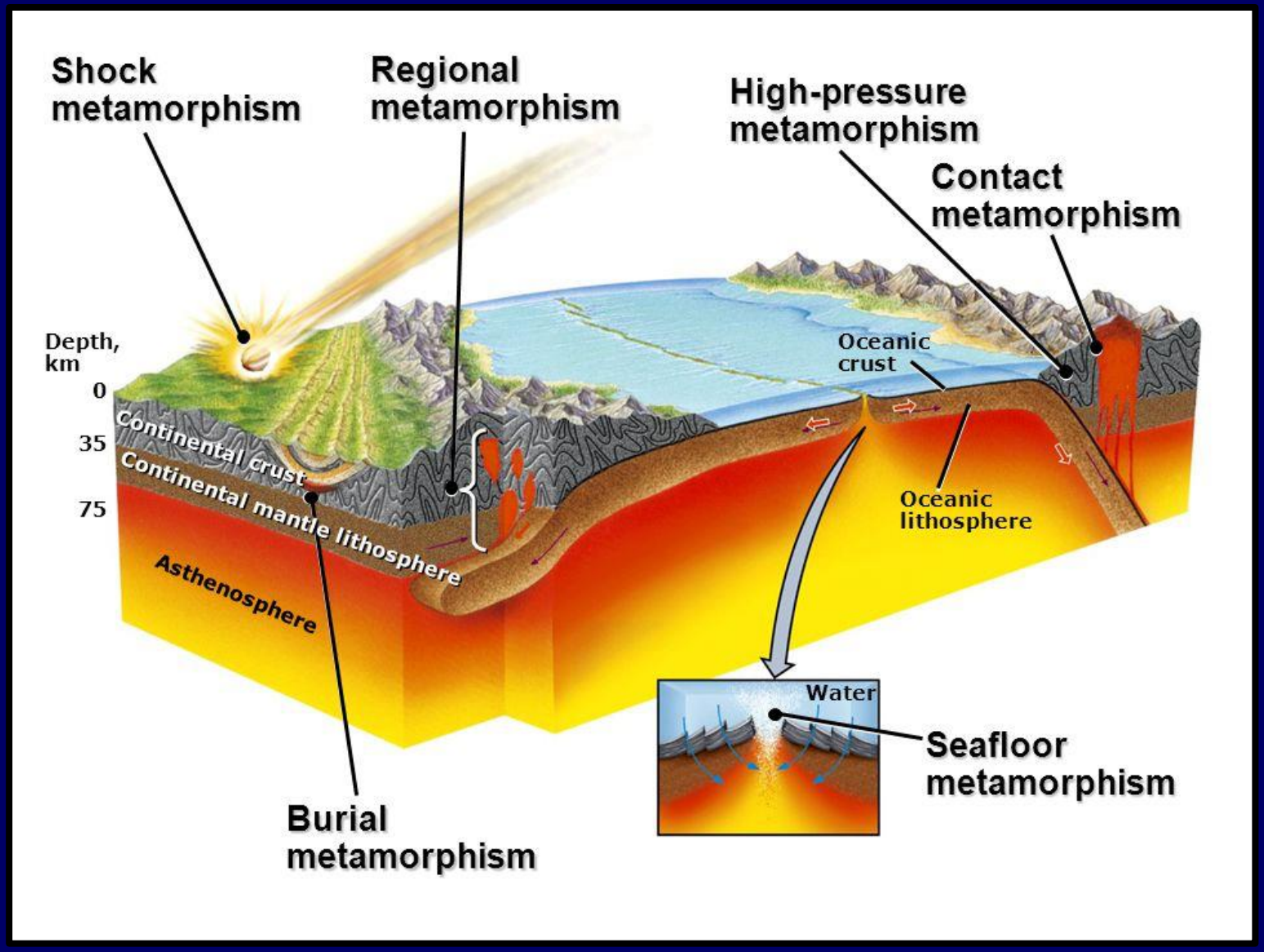


Quartzite

Metamorphism can cause chemical reactions



Types of metamorphism/location of metamorphism



Summary: Rocks

Rocks are naturally occurring, solid aggregates of minerals, or in some cases, non-mineral solid matter

Identity is determined by: texture and composition

Rocks are classified according to their origin, or genesis, into three types: igneous, sedimentary and metamorphic

The three families of rocks (igneous, sedimentary and metamorphic) are related through the Rock Cycle

The Rock Cycle is the result of interactions between the plate tectonic and climate systems

Intended Learning Outcome Revision

Now we have completed the class, you should be able to:

- explain the physical basis for the definition of a rock
- explain how you can identify different types of rocks on the basis of their texture and composition
- classify rocks according to their origin
- identify the geologic environments in which rocks form